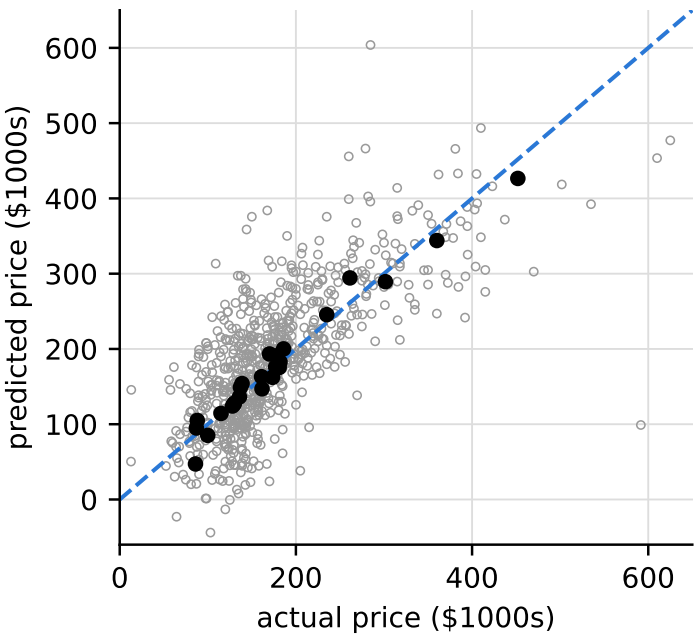
DATA 202 — Ridge and Lasso · Week 10 Wednesday

Sale prices of houses in Ames, Iowa, from a line on 18 columns that learned from only 25 houses. Your professor will tell you when to do each part.

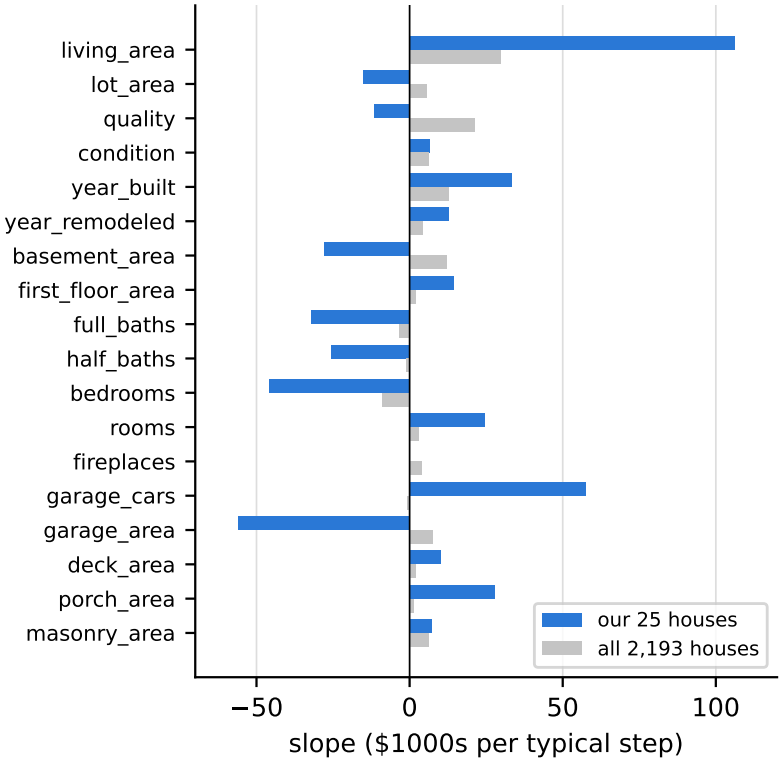
1 · Twenty-five houses



Each dot is a house: its actual price and the line's predicted price. On the dashed line, the prediction is perfect.

1. There are two kinds of dots: the **25 houses** the line learned from and **732 new houses**. **Label** each kind.
2. **Pair talk:** the line chose 19 numbers (18 slopes and an intercept) to fit 25 prices. Why does that make the 25 easy to fit? Why should we distrust the result?

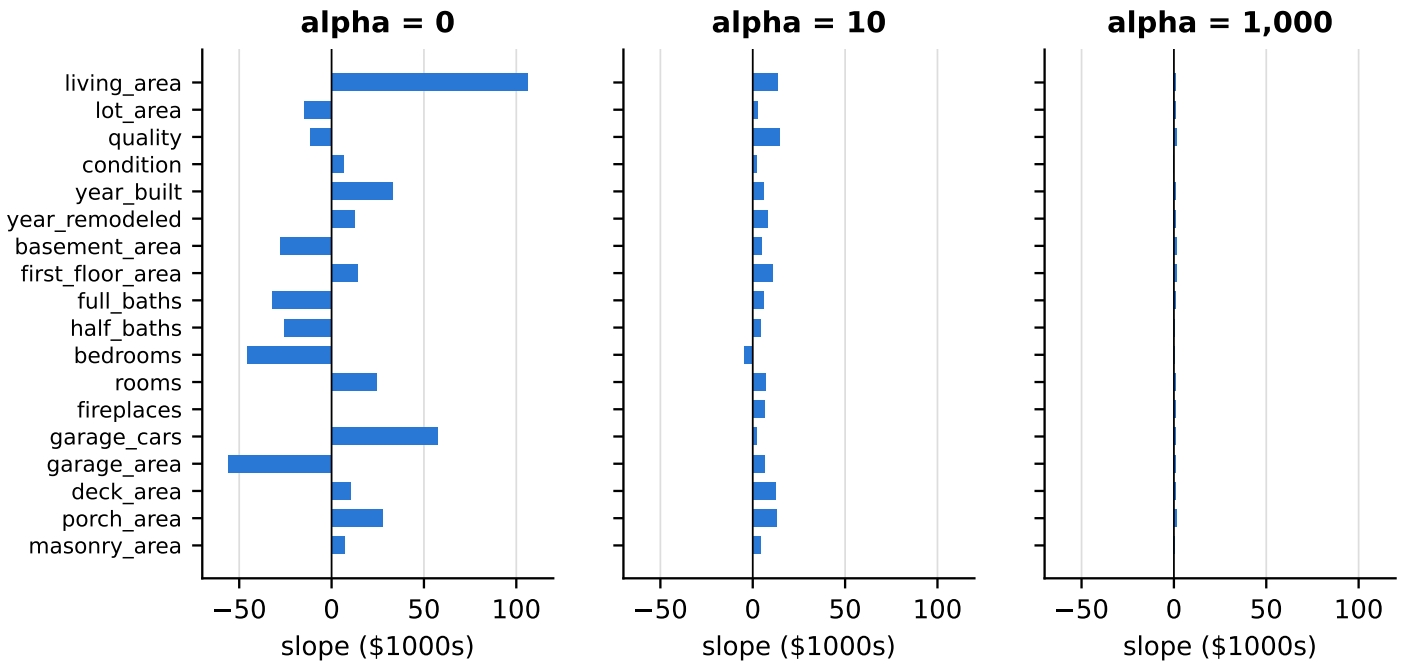
2 · Slopes



One slope per column, in dollars per typical step up in that column. Dark: learned from our 25 houses. Light: from all 2,193 training houses.

1. **Circle** three slopes that look too big or point the wrong way.
2. **Cross out** the columns that repeat another column.
3. **Pair talk:** `garage_cars` and `garage_area` got opposite slopes. Why might they cancel on the 25 houses but not on new ones?

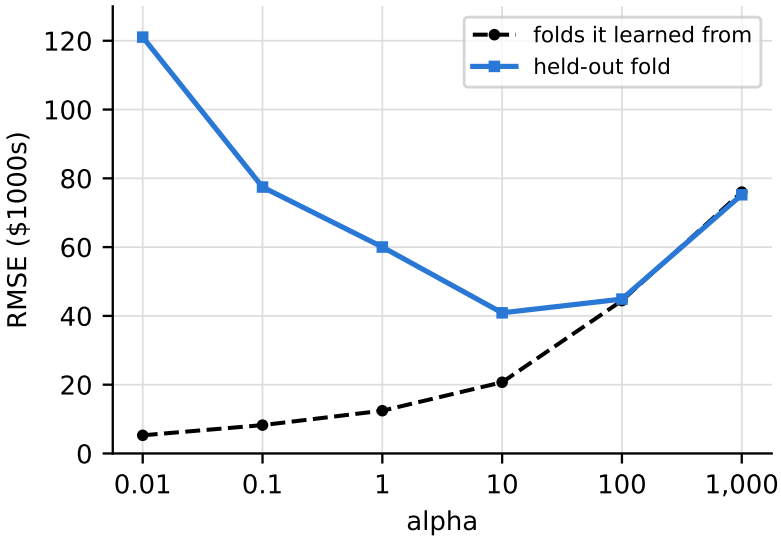
3 · Three alphas



Ridge's slopes on the same 25 houses, for three values of alpha. At alpha = 1,000 the bars are there, just short: about $1,000 each.

1. **Write** on each panel: *overfitting*, *about right* or *underfitting*.
2. **Pair talk:** with 25 *different* houses, which panel's slopes would change the most? Which would barely move? Is barely moving a good thing?

4 · Choose alpha



Ridge's typical miss in 5-fold cross-validation on the 25 houses.

1. **Circle** the best alpha.
2. **Label** the overfitting side and the underfitting side.
3. Here a *larger* alpha means a _____ model. For trees, a *larger* `max_depth` meant a _____ model.